

Review

Inclusive and Accessible Transportation for All: Strategies for Integrating Equity in Transportation Research

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Abstract: *Background:* Transportation equity has emerged as a critical area of focus in recent studies. Integrating equity into transportation research is increasingly recognized as essential for developing fair, inclusive, and sustainable transportation systems. The significance of this integration lies in its potential to reduce social and spatial disparities, support economic growth, promote social cohesion, and advance public health. *Methods:* This literature review synthesizes recent research on the incorporation of equity into transportation studies within transportation planning and frameworks. It examines how equity is conceptualized, measured, and operationalized, focusing on both horizontal equity (the equal treatment of similar groups) and vertical equity (prioritizing disadvantaged populations). This study evaluates various methodologies and tools, such as accessibility indices, affordability metrics, and community engagement approaches, which are used to assess and promote equity in transportation systems. *Results:* The findings from this review highlight the importance of both horizontal and vertical equity in transportation planning and policy and foundational tools for assessing equity. *Conclusions:* Integrating equity into transportation research is vital for creating systems that are equitable, inclusive, and responsive to the needs of all communities. Addressing current challenges requires clearer and more consistent frameworks, improved data collection, and more inclusive, community-driven decision-making processes.



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1. Introduction

Equity in transportation studies has received significant attention, as scholars and policymakers increasingly recognize the importance of creating fair and inclusive transportation systems. Traditionally, transportation research has primarily centered on issues such as infrastructure development, planning, and policy implementation. This conventional perspective has focused on improving the physical components of transportation networks and developing measures to improve their efficiency and effectiveness. The modern perspective frames transportation not only as a means of mobility and infrastructure but also as a tool for achieving social equity and justice. Although the intent of this study is to analyze how equity has been infused into transportation research, the existing literature does not adequately explore this topic in-depth. As a result, the aim of this study is to examine how equity considerations have been incorporated into transportation studies, specifically within planning and policy frameworks.

Studies have shown that transportation planning and policies often fall short in the aspect of incorporating equity, neglecting the needs of marginalized communities, including low-income groups and racial minorities [1–3]. This neglect has perpetuated and, in some circumstances, exacerbated existing social and economic disparities. These challenges have been highlighted in various studies, emphasizing the importance of incorporating social equity and justice considerations into transportation plans to ensure that all populations benefit from advancements in mobility and infrastructure.

One of the pivotal areas of transportation research is investigating how transportation policies and systems impact different demographic groups. Studies have shown that transportation inequities can limit access to essential services, such as healthcare, education, and employment, thereby perpetuating cycles of poverty and social exclusion [4]. Efforts have been made to address this gap in research by infusing equity into the transportation planning process and policies. Hickman and Banister [5] emphasized that transportation policies must consider the varied needs of different populations to avoid reinforcing social inequities. This viewpoint has gained popularity in recent years, with many researchers arguing for the inclusion of equity evaluations as an integral component of transportation planning procedures to ensure a fair distribution of benefits and burdens across all segments of society [2]. However, there have been variations in the way social equality objectives are defined, operationalized, and quantified in transportation planning. Additionally, there is a shortage of clear, practical ways for assessing the achievement of social equality objectives, emphasizing the need for enhanced procedures, such as multi-criteria decision-making approaches, to ensure that equity is properly integrated into transportation planning processes. By implementing such solutions, transportation design can include acknowledging the needs and rights of underrepresented populations while also actively working towards achieving a more equitable transportation system that serves varying population groups.

The role of public transportation as a tool for promoting social equity has been a central theme in recent research. Public transit systems, when designed and operated with equity in mind, can significantly enhance accessibility for underserved populations. Research has explored how Transit-Oriented Development (TOD) can contribute to more equitable urban growth. Loo et al. [6] examined how TOD can provide more affordable and efficient transportation options for low-income residents, thereby enhancing their access to jobs and services. Similarly, research by Fan et al. [7] highlights the potential of Bus Rapid Transit (BRT) systems to improve accessibility and reduce travel times for disadvantaged communities, suggesting that equitable investment in public transit can lead to more inclusive urban mobility networks. Recent studies have shown that infrastructure investments in these modes can yield significant benefits for disadvantaged communities [8–10]. However, the distribution of these investments has often been uneven [11]. These studies have seemingly demonstrated equity considerations in the transportation planning process but have not explicitly outlined where, how, and at what stages of the transportation planning process equity have been applied.

The increased emphasis on equity in transportation studies reflects a broader social recognition of the need for fair and inclusive public policies. As cities grow and people diversify, transportation systems must evolve, with equity being a top priority in research and planning projects. Ensuring equity in transportation planning requires a detailed examination of how different demographic groups are served by transportation networks, with an emphasis on meeting the needs of underserved communities. Equal access to transportation services and resources can be promoted by incorporating equity considerations throughout the transportation project's lifespan, from design to implementation.

This strategy not only promotes fairness, but it also adds to the general sustainability, inclusiveness, and efficiency of transportation networks.

To analyze how equity has been infused into transportation studies and the importance of transportation equity, this study aims to answer the following questions: (1) How has equity been infused in transportation studies over time? (2) At what stage in the transportation research process is equity not considered? (3) What measures or factors should be considered to ensure equitable transportation systems? To answer these questions, this literature review is structured as follows: Section two discusses in detail the concept of equity and the different types of equity. This provides a clear understanding of why equity is important in transportation research. Section three examines transportation planning and policies, detailing what transportation planning entails, which policies are implemented, and their impact on transportation systems, society, and individuals. Section four concludes the major findings, identifies gaps, and proposes recommendations for future research.

2. The Concept of Equity

Equity has been perceived to be a complex and multidimensional concept that has been actively debated for several decades in disciplines such as education, health, social sciences, and public policy [12,13]. Due to its intricate nature and the diverse contexts in which it is employed, a universally acknowledged definition for equity has not been established, despite repeated attempts to identify one. The intricacies and aspects of equity, which frequently encompass fairness, justice, and customized strategies for various requirements, add to the challenge of establishing a single definition. Equity draws attention to the differences and outcomes experienced by various individuals and groups and encourages the consideration of the rights, experiences, and benefits that each group may have [14].

Equity is a crucial factor in public policies and economics, as it heavily influences decisions that affect societal welfare. The concept frequently overlaps with matters of social justice, highlighting the need to tackle systematic disparities that place certain people at a disadvantage. Equity differs from equality, as it refers to a fair distribution that considers individual needs and circumstances, rather than simply aiming for uniformity [15]. Generally, equity, as opposed to other approaches, seeks to ensure that individuals are given the necessary resources and support to attain comparable results, thereby acknowledging and rectifying fundamental inequalities. This strategy is crucial in promoting a more inclusive and equitable society, guaranteeing that marginalized groups receive sufficient assistance to overcome systemic obstacles [16]. However, achieving equity faces significant challenges due to structural inequalities, implicit biases, and systemic discrimination. Kendi [17] discusses how entrenched racial biases in institutions and policies perpetuate inequities, while economic constraints and political resistance can impede the implementation of equitable policies. Promoting equity requires sustained, comprehensive efforts, including intersectional approaches that consider multiple dimensions of identity [18] and community engagement to ensure that marginalized voices are considered in decision-making processes.

To facilitate the incorporation of equity into research studies, particularly in the field of transportation research, efforts have been made to classify and define various aspects of equity. Many attempts have been made to classify equity based on various theoretical frameworks and practical applications. These classifications often draw from different schools of thought, such as utilitarianism, which emphasizes maximizing overall happiness, and egalitarianism, which focuses on ensuring the equal distribution of resources. These frameworks provide a diverse lens through which equity can be understood and

operationalized in policymaking and practice, each bringing its assumptions and implications for achieving fairness and justice in societal contexts [4]. In transportation research, the most common categories of equity addressed are horizontal equity, which involves egalitarianism and fairness, and vertical equity, which considers mobility needs and ability, income, and social class.

2.1. Horizontal Equity

Horizontal equity, sometimes known as fairness, equality, or egalitarianism, is a fundamental aspect of transportation equity. This notion is strongly based on the idea of ensuring an equitable and just distribution of transportation resources and costs among people or groups with similar needs and capabilities. At its core, horizontal equity advocates for the creation of a system in which users of transportation services enjoy benefits and amenities in proportion to their respective contributions. This approach promotes the idea of a reciprocal connection in which individuals take part in a fair exchange of resources within the transportation network. By adhering to this principle, the goal is to create an environment in which all members of society, regardless of their socioeconomic status or demographic characteristics, have equitable access to transportation services and endure a proportional share of the associated costs [19,20].

Mitigating externalities, such as transportation costs, is an important aspect of achieving horizontal equity. These external costs, such as traffic congestion and pollution, can have a major impact on communities and individuals who are not directly involved in transportation systems. By recognizing and managing these externalities, transportation systems can be designed using a more egalitarian framework in which the weight of such costs is fairly distributed among all people.

One way to completely consider and incorporate equity in transportation studies is by assessing horizontal equality through an extensive examination of how transportation resources, services, and impacts are divided across different groups. This involves exploring various geographic regions, income levels, and demographic groups that have similar mobility demands and capacities [21]. By conducting these extensive assessments, researchers can determine whether transportation networks are providing fair and equal access to all people. The examination of horizontal equity is a comprehensive procedure that uses modern approaches to ensure a precise analysis. Spatial analysis, for example, examines the spatial distribution of transportation services and resources to identify places that are underserved or overburdened, allowing for targeted improvements. Accessibility modeling evaluates how easily different populations can reach essential services and opportunities such as jobs, schools, and healthcare facilities, considering factors like travel time, cost, and convenience, providing a holistic view of transportation accessibility [20].

Horizontal equity has been integrated into transportation policy and practices in a variety of ways. For example, many cities have enacted policies to ensure equal service coverage in both wealthy and less affluent neighborhoods. This can be seen in public transportation systems that extend routes and boost service frequency in underserved areas. Furthermore, communities have used congestion pricing techniques to reduce traffic in congested regions while using funds to support transportation investments in places without adequate services. In doing so, they address the externalities of congestion and pollution, ensuring that the costs are distributed more equally among the population. Additionally, comprehensive bike-sharing programs have been established, with stations strategically placed to serve both high-density commercial zones and residential areas, ensuring equitable access to alternative transportation modes [22]. Despite these efforts,

there are still gaps in attaining horizontal equity, as discrepancies in access and service quality exist across areas and within populations.

Achieving horizontal fairness in transportation networks involves addressing several significant challenges. One primary issue is the persistent disparity in service quality and frequency between affluent neighborhoods and underserved areas, which often lack sufficient coverage [2]. This inequity is further exacerbated by the tendency to under-value external costs, such as traffic congestion and pollution, which disproportionately affect lower-income communities and communities of color. Additionally, limited data availability and the need for more advanced assessment methods complicate the measurement and analysis of equity [20]. The needs of non-car users are frequently overlooked, especially in suburban and rural regions, due to an insufficient focus on multi-modal integration [23]. Furthermore, disparities in funding and investment allocation can worsen these inequalities, as resources are often directed towards high-profile projects in wealthier areas. To address these challenges and move towards true horizontal equity in transportation, it is crucial to revise existing policies, improve data collection methods, and ensure inclusive decision-making processes that consider the diverse needs of all community members.

2.2. Vertical Equity

Vertical equity, embracing principles of social justice and inclusion, focuses on the distribution of resources among individuals and groups with diverse needs and capabilities. It is concerned with ensuring that disadvantaged groups receive a higher share of resources to compensate for broader societal inequities [19,24]. This form of equity is further divided into two main dimensions, which are discussed below.

2.3. Vertical Equity: Inclusivity and Accommodation of Mobility Needs and Abilities

Creating multimodal transportation systems that meet a variety of mobility needs is an important component of vertical equity [3]. This includes providing accessible infrastructure, vehicles, and services for individuals with disabilities, as well as accommodation for those who cannot or do not drive, such as children, the elderly, and those who do not own a car. Universal design principles are essential here, ensuring that transportation facilities and services are accessible to all without requiring special adaptations. This inclusive strategy enables everyone to effortlessly navigate the transportation system.

To assess vertical equity in mobility needs and abilities, it is important to consider the availability, accessibility, and quality of transportation alternatives for various user groups. This also entails assessing the extent of accommodation and inclusion in transportation planning and decision-making processes. Prioritizing inclusion allows transportation systems to better serve different communities and ensure fair access to transportation services and opportunities. However, there are still inadequacies in transportation studies that address vertical equity. Research frequently fails to address the requirements of certain groups, such as those with disabilities, resulting in inadequate infrastructure and services [25]. Furthermore, many studies do not fully assess how well transportation systems align with various requirements, particularly in rapidly growing metropolitan regions where infrastructure development may not keep up with the growing population [26].

2.4. Vertical Equity: Social Class and Affordability and Consideration of Income Disparities

A crucial aspect of vertical equity in transportation involves addressing income inequalities and making transportation more accessible for low-income groups. Individuals and households with lower incomes often face significant challenges in accessing and affording transportation, which can greatly restrict their mobility and ac-

cess to essential services and opportunities. Essentially, vertical equity emphasizes social justice by ensuring that resource distribution benefits the most disadvantaged and vulnerable populations [20].

Ensuring that public transportation services are competitively priced is critical to achieving vertical equity. This involves providing low-cost solutions to guarantee that transportation costs do not prevent people from moving around. Fare discounts and subsidies, such as reduced fares for low-income individuals, serve to ease the financial burden experienced. Furthermore, subsidized transit passes and fare-capping systems also help make transportation more inexpensive [27]. Integrating affordable housing with adequate transportation connections is a key approach. Strategic location planning places affordable housing in areas with reliable and accessible transportation, enabling residents to easily access essential services, employment, and other opportunities. Transit-oriented development encourages building near accessible public transportation hubs, minimizing the need for automobile ownership and increasing transportation accessibility for low-income individuals [2]. These strategies demonstrate the exploration and recommendation of vertical equity in transportation systems, which aims to ensure that the most disadvantaged and vulnerable groups receive the benefits of improved resource allocation.

2.5. How Is Transportation Equity Measured?

Transportation equity has emerged with a significant emphasis in urban planning and policy research, with experts proposing a variety of approaches for measuring and assessing equity in transportation systems. The measuring process of equity consists of three components to ensure that all aspects of equity are considered, which are (a) the costs and benefits, (b) the identification of the population and social groups, and (c) the distributive principles or criteria for assessing a transportation service's moral suitability, compared to its existing or predicted distribution. Incorporating these components in transportation planning and policies will ensure that equity is fully considered in the various aspects of transportation system design. Even with set rules and policies, it is still challenging to adequately capture or measure equity in its entirety [28,29]. Depending on the component of transportation being researched, studies have evaluated and integrated equity using a variety of criteria and tools that were suitable for assessing transportation systems. These criteria include accessibility, affordability, service quality, and the distribution of benefits and burdens across different demographic groups [30].

An important step to adequately measure transportation equity is having reliable and detailed data on geography, socioeconomic factors, and infrastructure over time. Without such comprehensive data, efforts to address transport challenges often overlook the specific needs of disadvantaged groups. Allen and Farber [31] found that, although lower-income neighborhoods in Canadian cities often have relatively good transit accessibility, nearly one million low-income individuals still live in urban areas with poor access to public transit. Their research highlights that transportation poverty is especially acute in certain dense, low-income neighborhoods located away from main transit routes and in low-density suburban areas, emphasizing the need for targeted transit planning to address these spatial inequalities. To improve the accuracy and relevance of equity assessments, studies have recommended actively engaging community organizations, as these groups are well-positioned to identify local barriers and needs that may not be visible through conventional data collection [16]. In addition, Manaugh and El-Geneidy [32] argue for the adoption of multi-dimensional equity frameworks that consider overlapping disadvantages, which can help policymakers better target transit investments and develop more equitable transportation infrastructure. Together, these approaches underscore the

importance of robust data collection and community involvement in creating equitable transportation systems.

2.6. Costs and Benefits

Cost–benefit analysis (CBA) is a systematic method used to evaluate transportation projects by comparing their expected costs and benefits. This approach is essential in decision-making processes because it provides a comprehensive assessment of the effects of transportation investments, ensuring effective resource allocation. The advantages of transportation projects identified through CBA include various factors, such as safety improvements, which reduce traffic accidents and fatalities, thereby enhancing societal well-being and economic productivity [20]. Travel time savings boost productivity by reducing commute times for individuals and businesses. Enhanced infrastructure also lowers vehicle running costs, resulting in fuel and maintenance savings [33]. Additionally, increased route quality, which includes comfort, convenience, and reliability, increases the total benefits of transportation initiatives.

On the cost side, CBA evaluates the economic resources required for the design, implementation, and maintenance of transportation infrastructure throughout its lifecycle. This includes initial capital input, ongoing operating expenses, and long-term maintenance or replacement costs [34]. By thoroughly evaluating these expenses, CBA ensures that all economic implications of a project are recognized and considered in decision-making processes. Equity considerations are crucial in CBA to ensure that benefits and costs are fairly distributed among different population groups, particularly in varied socioeconomic contexts where access and affordability gaps may exist. CBA uses distributive principles to assess the impact on different parts of the population, promoting fairness in transportation planning and policy [35].

While CBA is a widely used method for assessing transportation projects by quantifying costs and benefits in monetary terms, it has been criticized for potentially overlooking important factors such as distributional impacts and broader socio-economic considerations [30,36]. This limitation has led to the use of multi-criteria analysis as an alternative approach to integrating equity considerations into transportation project appraisals. One main criticism of CBA is its focus on economic efficiency, which may not adequately address how project benefits and costs affect different demographic groups [30]. Additionally, CBA may struggle to incorporate long-term effects and intergenerational issues. For instance, the environmental impacts of a new highway project might be undervalued in a standard CBA due to the difficulty in monetizing future climate change effects. This limitation can lead to decisions that prioritize short-term economic gains over long-term sustainability [37]. To address these shortcomings, researchers have proposed various approaches to enhance CBA. One method involves considering wider economic impacts, such as agglomeration effects and labor market impacts, which are not always fully captured in traditional CBA [38].

CBA remains a valuable tool for transportation project appraisal, and its limitations in addressing equity considerations and broader socio-economic factors have led to the exploration of alternative and complementary methods. By integrating these approaches, decision-makers can strive for a more comprehensive evaluation that balances economic efficiency with social equity and long-term sustainability.

2.7. Identification of Population and Social Groups

To effectively integrate equity into transportation studies, it is crucial to examine the methods and frameworks used to identify and address the needs of various population groups. Equity assessments must pinpoint specific populations and social groups impacted

by transportation policies, including disadvantaged groups such as individuals with disabilities, low-income families, visible minorities, and recent immigrants. This identification process ensures that the unique needs and challenges of these groups are considered in transportation projects, which is crucial for creating inclusive transportation policies [25,39]. Various methods, such as spatial analysis and demographic studies, are frequently employed to identify these groups. Spatial analysis allows researchers to investigate the geographic distribution of transportation resources and access, revealing disparities in service provision and accessibility [32]. According to Grengs [40], demographic studies provide detailed information about the characteristics of different population groups, enabling the customization of transportation services to meet their specific needs.

Furthermore, equity in transportation studies often incorporates frameworks like the Environmental Justice (EJ) framework and the Capability Approach. The EJ framework focuses on the equitable distribution of environmental benefits and burdens, ensuring that no group disproportionately suffers from negative environmental impacts [41]. The Capability Approach emphasizes enhancing individuals' abilities to achieve their desired functions and opportunities through equitable access to transportation [39]. Despite these efforts, significant gaps remain in the incorporation of equity in transportation studies. One major gap is the lack of comprehensive data on the transportation needs and patterns of disadvantaged groups, which hinders the development of targeted policies. According to Lucas [25], this lack of data prevents a full understanding of the unique challenges faced by these groups. Additionally, there is a need for more robust evaluation methods to assess the equity impacts of transportation projects and policies. Many current assessments are limited in scope and do not fully capture the long-term effects on different population groups [16].

Achieving fairness in transportation studies requires regular monitoring and policy adjustments to address emerging disparities, ensuring that all demographic groups benefit from transportation enhancements. This process includes identifying and rectifying current inequities while also anticipating future challenges and opportunities to foster a more inclusive and equitable transportation system. It is essential to continuously gather and analyze extensive data, formulate targeted policies, and employ rigorous evaluation methods to fully understand the impacts on different population groups. Through these efforts, transportation projects can become more inclusive and equitable, ultimately meeting the needs of multifaceted communities.

2.8. Distributive Principles and Criteria

Distributional effects describe how the impacts of policies or initiatives are distributed across different societal groups. These effects consider factors like income, age, location, and mobility [42]. To ensure the fair allocation of benefits and burdens among various population groups, policymakers use distributive principles or criteria as guidelines. Distributive principles are strategies used to analyze the distributional effects of policies, ensuring that the benefits and burdens are fairly allocated among various populations [20]. These principles are crucial when analyzing the equity of transportation systems. Researchers have proposed multiple frameworks to assess transportation equity, incorporating various distributive equity principles. One influential approach combines Rawlsian theory with Capability Approaches (CA). This perspective, highlighted by Pereira et al. [26], emphasizes the importance of ensuring that all citizens have a basic level of access to essential destinations. This aligns with Rawls' theory of justice, which focuses on improving conditions for the most disadvantaged members of society [43].

Building on this concept, Martens [44] introduced the idea of an "accessibility poverty line", a threshold that helps identify individuals or areas lacking adequate transport-related

accessibility. This approach allows researchers and policymakers to focus on interventions that benefit those most in need of improved transportation access. Another crucial aspect of transportation equity analysis is the consideration of both horizontal and vertical equity. As previously highlighted, horizontal equity involves treating people with similar needs and abilities equally, while vertical equity addresses the distribution of impacts among groups with differing abilities and needs. In practice, this means evaluating whether transportation resources and benefits are distributed fairly across various socioeconomic groups and geographic areas.

To operate these principles, researchers have developed several metrics and indicators. For instance, Golub and Martens [45] introduced the “access ratio”, which compares how easily public transit users can reach destinations compared to car users. They proposed that transit users should be able to access at least one-third of the opportunities available to car users, establishing an “access poverty line”. To further assess equity, Di Ciommo [46] created an “inaccessibility index”. This tool measures how many desirable activities different groups of people cannot reach, helping planners evaluate how transportation decisions impact equity. Building on this work, Adli and Donovan [47] developed a “justice test” for transportation planning. This test examines how changes in accessibility affect various socioeconomic groups, giving planners a way to assess the fairness of proposed transportation projects or policies. These metrics provide practical ways for planners and policymakers to measure and improve transportation equity in different communities.

Transportation studies have made tremendous progress in evaluation methods, but concerns persist about the adequate integration of equity considerations. An examination of transportation plans across North America revealed a disconnect between stated social equity goals and their practical implementation, with many plans failing to translate objectives into concrete targets or to give them adequate consideration in decision-making processes [2]. Key challenges include inconsistencies in defining, operationalizing, and measuring social equity objectives, indicating the need for clearer, more actionable methods. To address these issues, the use of advanced methodologies like multi-criteria decision-making, increasing the importance of equity factors in project prioritization, and enhancing the involvement of disadvantaged groups in the planning process have been recommended [48]. While various frameworks and metrics have been proposed, their application remains inconsistent. As a result, the creation of standardized techniques to incorporate equitable issues into transportation policy and project evaluation are needed. Moving forward, transportation studies should prioritize developing more robust and universally applicable methods for integrating equity considerations, improving evaluation techniques, and devising innovative ways to involve underrepresented communities in shaping transportation policies and projects that affect their lives.

2.9. Consequences of Equity Incorporation

Policymakers and planners utilize a variety of quantitative metrics and statistics to integrate “fairness” into public transit. For instance, travel cost deprivation coefficient accounts for differences in travel expenses between various income groups [35], while the Gini coefficient evaluates the equity of resource allocation, such as road area or accessibility [26], and accessibility indices, which gauge how simple it is to get to jobs, services, or destinations using public transportation, and these are important metrics. Access to transportation, transportation affordability, and cumulative opportunity measures (the number of options attainable within a specific travel time) are additional pertinent variables. To guarantee that upgrades benefit disadvantaged communities and lessen current gaps, advanced methods, including bi-level programming approaches,

can directly incorporate these fairness restrictions into the planning and optimization of transport networks [49].

There are significant distinctions between deliberately integrating fairness into public transportation and not doing so. When equity is not taken into consideration, transportation innovations frequently favor already privileged groups disproportionately, increasing the accessibility and opportunity gap between high-income and low-income or marginalized groups. According to a study conducted in Shenzhen, China, by Zhou et al. [50], for instance, transportation improvements benefitted those with already better access; thus, while total accessibility increased with transit expansion, the gap between inhabitants of urban villages—who often have lower incomes—and nonurban villages also widened. To identify and address the requirements of transportation disadvantaged groups, equity research also necessitates classifying populations by demographic or geographic criteria, such as income level, car ownership, or residence location.

Transportation systems, on the other hand, can significantly lessen inequities when fairness is included, by decreasing deprivation coefficients and optimizing equitable resource allocation. A more socially inclusive transit system, better accessibility for low-income and vulnerable populations, and more balanced mode shares are the results of this. In the end, public transportation planning that incorporates fairness not only addresses socioeconomic equity but also improves the system's overall effectiveness and appeals to all users.

3. Transportation Equity Measuring Tools, Indicators, and Models

Transportation equity assessment tools and indicators are approaches for determining the fairness and distribution of transportation benefits and burdens among various population groups. These methods frequently use indicators like affordability, mobility, safety, and environmental effects to assess transportation systems. Some of these tools and indicators are discussed in detail below in Table 1.

Table 1. Analysis of transportation equity measuring tools, indicators, and models.

Transportation Research	Equitable Actions/Measures	Tools	Procedures	Effect/Outcomes	References	Limitations	Comment
Accessibility and Connectivity	Develop equity measures for public transit connectivity; compare connectivity across different socio-economic groups.	Connectivity metrics; GIS (Geographic Information Systems).	Analyze transit connectivity indices using GIS; assess disparities in accessibility.	Improved understanding of connectivity equity; identified areas with low connectivity affecting disadvantaged groups.	[51]	Limited to specific geographic areas; may not capture all aspects of accessibility	Focuses on the spatial distribution of connectivity; useful for identifying inequities in transit access.
	Measure equity using transit connectivity metrics; develop strategies for improving service in underserved areas.	Connectivity indices; data on transit routes and service frequencies.	Applied connectivity metrics to evaluate equity; develop recommendations based on findings.	Enhanced measures for evaluating equity; provided recommendations for improving transit service in underserved areas.	[52]	May not address all dimensions of equity; reliant on the accuracy of connectivity data.	Provides actionable recommendations for enhancing equity in transit provision; focuses on service improvements.
	Implementation of pro-poor public transport subsidies; focus on affordability and accessibility improvements for low-income populations.	Statistical analysis (regression analysis, descriptive statistics); accessibility and affordability metrics; and survey data.	Regression analysis to evaluate the relationship between subsidies and measures of accessibility and affordability; use of descriptive statistics to summarize survey data and measure the impact of subsidies on low-income users.	Improved accessibility and affordability for low-income groups; identified gaps in subsidy coverage and effectiveness.	[53]	Context-specific to Bogotá; may not be generalizable to other regions with different economic and social conditions.	Provides a detailed assessment of the effectiveness of targeted subsidies in improving transport access and affordability for low-income residents in an urban setting.

Table 1. Cont.

Transportation Research	Equitable Actions/Measures	Tools	Procedures	Effect/Outcomes	References	Limitations	Comment
Transportation Infrastructure	Use accessibility measures to evaluate social equity in public transport provision.	Accessibility analysis; GIS.	Measured accessibility to public transport before and after infrastructure projects.	Improved understanding of which populations benefit from new infrastructure.	[32]	Focuses mainly on public transport; may not cover all modes of transportation.	Highlights the need for accessibility measures in planning for equitable outcomes.
	Incorporate equity considerations in transport infrastructure evaluation.	Multicriteria analysis (MCA); Cost–Benefit Analysis (CBA).	Integrate equity criteria into evaluation frameworks; propose new methodology.	Enhanced evaluation processes; better inclusion of equity considerations.	[36]	The methodology may be complex to implement and requires comprehensive data.	Proposes a new approach for systematically including equity in infrastructure evaluation.
	Incorporate social equity in infrastructure planning and delivery.	Web-based tool; Equity Assessment Tools.	Developed and implemented a web-based tool to help planners incorporate social equity considerations in the planning and delivery of transportation infrastructure.	Enhanced ability of planners to consider and incorporate social equity in their decision-making processes; improved infrastructure outcomes for disadvantaged groups.	[54]	Dependence on the accuracy and comprehensiveness of the data input into the tool; potential user bias in interpreting tool outputs.	Demonstrates the practical application of technology in enhancing equity considerations in infrastructure planning and delivery.

Table 1. Cont.

Transportation Research	Equitable Actions/Measures	Tools	Procedures	Effect/Outcomes	References	Limitations	Comment
Transportation Planning	Plan for transportation equity in small regions; developing meaningful performance assessments.	Performance assessment tools; GIS.	Established performance measures that reflect equity considerations and apply these measures to small regional transportation plans.	Improved identification and understanding of equity issues in small regional transportation planning; more targeted and effective interventions.	[55]	Limited scope to small regions, may not be directly applicable to larger urban areas; relies on availability and accuracy of local data.	Provides valuable insights for addressing transportation equity in smaller regions' highlighting the importance of context-specific performance measures.
	Systematically incorporating equity considerations in transportation planning.	Equity evaluation frameworks, guidelines; performance metrics; and indicators.	Reviewed existing literature and practices to propose a comprehensive framework for including equity in transportation planning.	A more structured approach to transportation planning that ensures that equity considerations are systematically included, leading to more equitable transportation outcomes.	[14]	Implementation challenges due to varying definitions and interpretations of equity, potential resistance from stakeholders to adopt new frameworks.	Emphasizes the need for a comprehensive and systematic approach to integrating equity in transportation planning to achieve fairer outcomes.
	Critical evaluation of equity objectives and measures in urban transportation plans in North America.	Equity objectives and measures evaluation; content analysis; comparative analysis.	Analyzed transportation plans to evaluate how well they integrate social equity considerations and identify gaps and areas for improvement.	Enhanced integration of social equity in transportation plans; identification of best practices and common shortcomings.	[2]	May not account for local contextual differences; reliance on the availability of detailed transportation plans for analysis.	Provides a comprehensive critique and recommendations for improving the integration of equity in urban transportation planning.

Table 1. Cont.

Transportation Research	Equitable Actions/Measures	Tools	Procedures	Effect/Outcomes	References	Limitations	Comment
Environmental Transportation Studies	Reduce exposure to particulate air pollution during active travel to promote health in urban planning.	Air pollution models; GIS; health impact assessments.	Analyzed air pollution exposure levels associated with active travel; integrated low-exposure planning into urban design.	Improved planning strategies to reduce air pollution exposure during active travel, leading to healthier urban environments and increased public health benefits.	[56]	Data may be limited by the availability of air quality measurements; model accuracy may vary.	Emphasizes the importance of integrating air quality considerations into urban planning to enhance health outcomes for active travelers.
	Address equity concerns in protecting transportation infrastructure from sea level rise.	Sea level rise models; equity assessment tools (e.g., vulnerability assessments).	Evaluating the impact of sea level rise on transportation infrastructure; identifying vulnerable communities; proposing equity-focused protection measures.	Enhanced understanding of the equity implications of sea level rise on transportation infrastructure, leading to more equitable adaptation and protection strategies.	[57]	May face challenges in accurately predicting future sea level rise impacts and the variability in infrastructure vulnerabilities across different regions.	Emphasizes the need for equitable infrastructure protection to safeguard vulnerable communities from sea level rise.
	Incorporate health and equity considerations into transportation health impact assessments (HIAs).	Health impact assessment tools; equity metrics (e.g., disparity indices).	Retrospectively analyzing HIAs to assess the integration of health and equity considerations and proposing improvements to enhance their effectiveness.	Improved methodologies for incorporating health and equity considerations in HIAs, leading to more comprehensive and equitable transportation planning outcomes.	[58]	Analysis may be limited by retrospective data; the effectiveness of proposed improvements depends on their adoption by practitioners.	Provides valuable insights into improving the integration of health and equity in HIAs, leading to better-informed and fairer transportation planning decisions.

Table 1. Cont.

Transportation Research	Equitable Actions/Measures	Tools	Procedures	Effect/Outcomes	References	Limitations	Comment
Transportation Affordability	Address transport poverty by analyzing accessibility issues for low-income households in Canada.	Accessibility measures; transport poverty indicators; national data analysis tools.	Assessed inaccessibility levels for low-income households using national data; recommended policies to improve access and affordability.	Enhanced understanding of transport poverty among low-income households, leading to targeted policies for better accessibility and affordability.	[31]	Limited by data availability and variability in local contexts may not capture all nuances of transport poverty.	Identifies the necessity for specialized interventions and policies to address the specific needs of low-income households in transportation planning.
	Evaluate transportation affordability using a context-sensitive framework.	Affordability frameworks; cost–benefit analysis; context-sensitive evaluation tools.	Applied a context-sensitive framework to evaluate transportation affordability across various contexts and demographic groups.	Improved technique for assessing transportation affordability, resulting in context-sensitive policy recommendations and planning decisions.	[27]	The framework may not account for all local variables and economic factors; it may need to be adapted for different circumstances.	Provides a comprehensive framework for assessing transportation affordability, emphasizing the importance of context in evaluating affordability across diverse populations.

Table 1. Cont.

Transportation Research	Equitable Actions/Measures	Tools	Procedures	Effect/Outcomes	References	Limitations	Comment
Transportation Safety Analysis	Explore equity in non-motorist safety by identifying disparities and addressing safety for pedestrians and cyclists in Houston.	Safety analysis tools; GIS; equity metrics (e.g., Safety Disparity Index, Exposure to Risk, Access to Safety Infrastructure).	Analyzed safety data for non-motorists (pedestrians and cyclists), mapping high-risk areas and evaluating equity in safety interventions.	Improved understanding of safety disparities for non-motorists; targeted interventions in high-risk areas to enhance safety for underserved communities.	[59]	May be limited by the availability and accuracy of safety data; potential challenges in generalizing findings to other regions.	Highlights the importance of addressing safety disparities and targeting interventions to improve non-motorist safety in underserved areas.
Traffic Management	Evaluating equity in traffic safety issues identified proactively vs. reactively.	Traffic safety data analysis tools; equity assessment frameworks; GIS tools.	Analyzing traffic safety data to compare equity outcomes of proactive vs. reactive identification of safety issues.	Findings suggest that proactive identification of safety issues can lead to more equitable safety improvements.	[60]	Data limitations may affect the generalizability of results; proactive measures may require higher initial investments.	Provides insights into the benefits and limitations of proactive versus reactive identification of traffic safety issues, emphasizing the need for equitable safety measures.

This table summarizes selected studies on transportation equity, in diverse research areas. It outlines equitable measures, tools, procedures, outcomes, limitations, and key insights. It shows how different methods and data sources are used to evaluate and promote equity. All data used in this table are sourced from peer-reviewed research and reliable databases.

3.1. Accessibility

Accessibility is a key component in determining transportation equity since it refers to how quickly people can reach vital destinations such as schools, workplaces, and healthcare facilities. This term extends beyond distance to include the quality and convenience of various transportation options. The issue of accessibility has been viewed as a method-specific measure that is used for various mobility concerns [61]. It is measured using various tools, depending on the perspective by which the issue of accessibility needs to be addressed in transportation studies. For instance, Geographic Information Systems (GIS) have become invaluable tools for measuring accessibility, offering spatial analysis to assess the connectivity of transportation networks [62]. Accessibility also considers the inclusiveness of transportation modes, which ensures that transportation services are accessible to individuals, including those with disabilities. Various metrics such as the roadway level of service, congestion delay, and travel speeds have been used to gauge how well transportation systems serve diverse populations [20]. These metrics quantify the efficiency and efficacy of transportation networks, emphasizing areas for improvement.

A range of methodologies have been explored to assess accessibility and equity in transportation. One such method involves using Lorenz curves to evaluate public transport equity by comparing accessibility distribution across different population groups [63]. Another method for evaluating the equitable distribution of transportation facility availability is the use of connectivity measures, which consider how well public transportation connects various locations [52]. These methods have been applied in various real-world scenarios. For instance, in Bogotá, an assessment of ‘pro-poor’ public transport subsidies demonstrated significant accessibility improvements for low-income groups [53]. Similarly, a study in Melbourne employed a clustering method to identify areas with notable disparities in public transportation access [64]. Accessibility metrics, such as the number of jobs available within a given travel time via various modes of transportation, are critical for assessing transportation equity. These measures consider not only physical infrastructure but also the connectivity and integration of other modes of transportation, such as walking, cycling, using public transit, and driving. In their study, Aman et al. [65] explored the spatial accessibility of micro-mobility services, such as bikes and scooters, among disadvantaged communities. The study showed that the use of the Lorenz curve and spatial regression models aided in identifying significant inequities in access to bikes and scooters, particularly for transit-dependent and black populations, compared to other groups. These findings highlight the need for adequate micro-mobility planning to enhance transportation equity across disadvantaged groups.

Additionally, equity-focused metrics such as the Gini coefficient and Lorenz curve can be applied to transportation studies to assess how benefits and burdens are distributed among different socioeconomic groups [53]. These measures help to identify inequalities in access and service quality, guiding efforts to create a more inclusive transportation system. Combining traditional transportation metrics with equity-focused measures provides a comprehensive understanding of how well the transportation system performs and serves all communities, informing policies and interventions aimed at improving access and mobility for everyone.

Despite progress in measuring accessibility and equity, challenges persist. One major challenge is the need for comprehensive data that accurately represents the travel patterns and needs of diverse population groups. In addition, there is growing recognition of the need for dynamic accessibility measures that account for temporal variations in transportation service availability [66]. Research centered on developing more advanced tools and methodologies capable of capturing the intricacies of transportation systems and

their impacts on different communities is needed to continue making advancements in the transportation equity space.

3.2. Affordability

Affordability in transportation tackles the financial burden that transportation costs place on individuals and households, disproportionately affecting low-income populations. High transportation costs can limit access to critical services and opportunities, highlighting the importance of effective monitoring and policy responses. Affordability analysis identifies biases in transportation planning and finance that favor more expensive modes over more inexpensive ones, allowing targets to be set to improve affordable travel options [20]. The Location Affordability Index (LAI) is one of the measures used to analyze affordability. The LAI assesses housing and transportation costs as a percentage of household income, providing a comprehensive measure of economic pressure [67]. This aids in identifying financial burdens and informing measures targeted at mitigating these challenges. The Housing + Transportation (H + T) Affordability Index is another important metric in evaluating affordability. This index assesses the combined costs of housing and transportation relative to income, offering insights into the total financial impact on households [68]. The Transportation Cost Index (TCI) is also an essential tool used to evaluate transportation affordability with household income, focusing on the financial implications of different transportation modes and services as well as land use systems [69].

Affordability is intricately linked to accessibility, as financial barriers can significantly limit an individual's ability to use transportation services effectively. Ensuring that transportation remains affordable across various economic levels is crucial for equitable access. This involves examining the availability of affordable transportation options, such as subsidized public transit, biking infrastructure, and pedestrian pathways. Equitable transportation could be enhanced by promoting policies like fare subsidies and discounted transit passes [53].

The relationship between housing and transportation costs is substantial, as both expenses frequently account for about half of the average US household budget. Tools like the LAI and H + T Index emphasize the significance of considering both costs when determining true affordability. Liao et al. [68] highlighted the importance of including housing and transportation affordability in long-term spatial planning. Similarly, this underlines the importance of transportation cost regulations in promoting equity for low-income groups. To understand the complex relationship between housing and transportation costs, it is essential to determine the efficacy of a multi-modal approach across diverse demographics and areas, as well as their long-term effects on community growth and sustainability [70,71].

3.3. Service Quality

Service quality is a critical element in evaluating transportation equity, encompassing several dimensions such as frequency, reliability, safety, and comfort of transportation services. These factors play a significant role in ensuring equitable access, as substandard service quality can disproportionately impact disadvantaged communities. Typically, surveys and user feedback are utilized to assess service quality, shedding light on the user experience and pinpointing areas that need improvement [2,72]. In assessing service quality, various metrics are considered, including on-time performance, vehicle condition, and passenger satisfaction. High-quality transportation services are essential for achieving equity in transportation, as they enable all users—regardless of socioeconomic status—to access reliable and efficient services. Enhancing service quality can lead to increased ridership, shorter travel times, and improved overall satisfaction, thereby contributing to a more equitable transportation system [72,73].

The integration of user feedback and performance metrics into service quality evaluations offers a thorough understanding of a transportation system's strengths and weaknesses. This approach ensures that the needs of all community members, particularly those from underserved populations, are accounted for in transportation planning and policy decisions [72]. Despite the importance of service quality, certain areas require more attention. For example, while surveys and feedback are useful, they often capture only a snapshot of the user experience and may not fully reflect the long-term or nuanced impacts of service quality on different population groups. Additionally, the impact of service quality on accessibility outcomes for individuals with disabilities or other specific needs may not be adequately addressed in current evaluations.

Developing more comprehensive metrics that capture these dimensions of service quality and their implications for transportation equity is necessary. Additionally, understanding the differential impact of service quality improvements across various socio-economic and demographic groups remains underexplored, which could offer deeper insights into how to effectively target improvements to benefit all segments of the population equitably. By addressing these areas, researchers can ensure that transportation systems are meeting the needs of diverse population groups.

3.4. Transportation Equity Models

Transit agencies and researchers employ different models and design approaches to evaluate the degree and methodology of equity integration within transportation planning initiatives. The goal of this thorough assessment process is to quantitatively analyze how equity considerations have been incorporated into various phases of planning and implementation, such as grouping people based on socioeconomic status, assessing equity measures for each group, and qualitatively analyzing how minority and low-income communities have been mapped near proposed transportation projects to identify disproportionate impacts like traffic and pollution [74,75].

One of the most frequently used transportation equity models is the travel demand model. Numerous research efforts have utilized travel demand models to evaluate the fairness of transportation initiatives and regulations. Essentially, these models are employed to predict alterations in travel patterns due to specific transportation projects or policies. Some common travel demand models that have been used and described in studies include the activity-based travel demand models, the integrated transportation land use and microeconomic models, and travel demand forecast models, such as the microsimulation model [74].

Bills [76] described how the San Francisco County Transportation Authority used an advanced, tour-based travel demand microsimulation model for equity analysis in transportation planning. This approach, which operates at the individual level, outperforms existing models by identifying groups of concern based on income, car availability, household composition, and gender. It assesses equity using mobility and accessibility metrics such as travel time and job access, which results in a more precise assessment of transportation policy consequences. In another study by Sall et al. [77], a Nine-County Regional Pricing Model (RPM-9), an enhanced version of the San Francisco Chained Activity Modeling Process (SF-CHAMP) travel demand model, was also used by the San Francisco County Transportation Authority (SFCTA) to evaluate congestion pricing strategies in San Francisco. This model allowed for a detailed analysis of impacts across different population segments, incorporating value-of-time distributions and enhanced peak spreading models. It was used to analyze various discount programs aimed at addressing equity concerns. These models have proven to be effective by ensuring that equity is considered at every level in transportation projects. In their study in Perth, Australia, Ricciardi et al. [1] assessed

the use of travel demand and accessibility models to explore public transportation equity among various disadvantaged groups. Their research highlights how such models can identify spatial and socioeconomic disparities in transit access, supporting more equitable policy interventions. Similarly, in Europe, the Bureau and Glachant [42] analyzed the distributional effects of public transport policies in Paris, employing quantitative models to evaluate how changes in fares and service modifications impact different population groups. Their findings highlight the importance of integrating equity considerations at multiple stages of transportation planning, from policy design to implementation.

Another model is the Justice-and-Rights-Based transportation equity model. This model is drawn from foundational theories of distributive justice and the notion of rights to guide both the assessment and design of transportation systems. Studies have assessed this area, advancing frameworks that move beyond traditional cost–benefit or demand-based analyses by explicitly asking whether transportation resources and opportunities are distributed fairly among all population groups [14,39,44]. Martens [39] applied Walzer’s “spheres of justice” to the transportation sector, arguing that access to transportation and, by extension, to opportunities in society should be treated as a distinct social good, governed by its principles of justice. This perspective is further developed in another of their studies [14], emphasizing that transportation planning should prioritize improving accessibility for those who are worst off, thus operationalizing “Rawlsian principles of justice”. In their study, Golub and Martens [45] applied these ideas in practice by evaluating the modal equity of regional transportation plans, examining whether policies and investments close gaps in accessibility for marginalized or disadvantaged groups, rather than simply maximizing aggregate benefits.

Building on these normative foundations, justice-based approach models have led to the development of practical assessment tools for policy and project evaluation. For instance, Nazari and Donovan [47] introduce “Justice Tests” for public transport investments, applying the “Right to the City” concept to systematically evaluate whether proposed projects promote a fair distribution of benefits and mitigate negative impacts. Their framework encourages planners to consider not only the spatial distribution of transport services but also the social and economic implications for vulnerable populations, such as low-income or minority communities. By embedding justice and rights considerations into the planning process, these models aim to ensure that transportation systems do not perpetuate or exacerbate existing inequalities but instead serve as tools for greater social inclusion and equity.

Despite progress in transportation equity models, there are still limitations. This includes the need for more accurate estimates of disenfranchised groups, difficulties in successfully explaining complex scenarios to the public, and the awareness that properly resolving equality concerns may necessitate studies beyond the travel demand model’s capabilities. Additionally, the models were limited in their features and lacked important policy benefits. To address these limitations, research should focus on improving the precision of demographic estimates, strengthening public communication tactics, and including a broader variety of policy consequences to provide a more thorough analysis of equity concerns.

4. Transportation Planning

Transportation planning is a multifaceted field that synthesizes knowledge from various disciplines, such as ecology, computer science, architecture, communications, public policy, and other related fields. This integration aims to prepare prospective planners with the requisite analytical competencies to effectively address pressing mobility challenges [78]. Transportation planning plays a critical role in the broader context of transportation project

development by evaluating transportation alternatives, applying system engineering principles to complex networks, and using mathematical modeling and computer-aided analysis for decision-making processes [79]. This dynamic curriculum reflects the transportation planning sector's increasing recognition of the need for adaptable professionals who can adeptly navigate emerging trends, technological advancements, and complex challenges within the field.

The purpose of transportation planning is to develop strategies that ensure that mobility is accessible to all members of society, with a particular focus on addressing inequities in transportation systems. These planning efforts are particularly important in mitigating the impacts of environmental concerns, new technologies, economic factors, and inequities that have significantly impacted society and individuals [21]. Equity concerns in transportation planning revolve around the differential impacts and consequences of planning projects and system developments on historically underserved communities [14]. For example, the implementation of the Federal-Aid Highway Act of 1956 led to the extensive development of interstates and highways, displacing approximately 475,000 households and one million individuals from low-income communities and minority groups [80,81]. Additional research studies also prove that there are instances where infrastructure developments have produced inequitable outcomes both domestically in the US and globally [82,83]. This further emphasizes the need for proactive measures to ensure that future planning developments do not persist in perpetuating inequities across vulnerable communities.

5. Transportation Policies

To address the adverse impacts of transportation projects on underserved communities, various policies have been implemented to foster more equitable infrastructure and transportation systems. These policies include, but are not limited to, congestion pricing, transit-oriented development (TOD), and Vision Zero.

5.1. Congestion Pricing

Congestion pricing was developed as a policy response to address the increasing traffic volumes that exceeded the capacity of existing transportation infrastructure. It aims to improve traffic flow and reduce environmental impacts by implementing additional charges for private vehicle usage during peak demand periods [84]. Initially proposed by Professor William Vickrey of Columbia University in 1952 to manage the New York City subway system, congestion pricing uses variable fare structures based on factors such as the time of day, direction of travel, and real-time traffic conditions to influence travel behavior. This approach aims to modify travel behavior by encouraging alternatives to private transportation or discouraging travel during peak periods [85], thereby fostering equitable transportation.

Congestion pricing policies have been implemented across metropolitan cities and regions globally, including London, Singapore, Stockholm, and certain areas within the United States to enhance safety measures [86]. However, the policy's implementation raises equity concerns, particularly regarding its impact on low-income residents who may face financial constraints when using priced routes. This has implications for transportation equity, as it can limit mobility for those who cannot afford the additional costs. To address the issues posed by congestion pricing, a mechanism such as revenue recycling is employed. The term "revenue recycling" refers to a mechanism whereby income generated from policies is allocated and returned to society. In the context of congestion pricing policies, the primary purpose of revenue recycling is to mitigate the financial burden imposed by the congestion pricing system on vulnerable populations, while simultaneously pursuing initiatives to improve the economy [87]. Although revenue recycling promotes equitable

outcomes, some researchers have contended that it is regressive and benefits individuals at higher socioeconomic levels, particularly when such mechanisms are employed to reduce income or corporate taxes, which typically confer greater advantages to those with elevated tax brackets [88]. Another theorized argument within this discourse is that those of lower socioeconomic levels, in contrast to their more affluent counterparts, are disproportionately subject to congestion-priced levies due to their limited flexibility in determining work schedules and commuting distances [89]. To ensure the equitable distribution of benefits from revenue recycling across all socioeconomic sectors, policies such as congestion pricing should be implemented as progressive redistribution strategies. Budolfson et al. [90] asserted that this approach is designed to allocate a larger proportion of benefits to lower-income groups, thereby counterbalancing the advantages that higher-income groups may already derive from such policies.

5.2. Transit-Oriented Development

Transit-oriented development (TOD) is an urban planning and design approach that optimizes the utilization of residential, commercial, and leisure spaces within the proximate walking distance of public transit [8]. Although TOD has origins in earlier housing design concepts, urban planner Peter Calthorpe was the first to reintroduce and formally establish the concept in his influential publication, *The Next American Metropolis: Ecology, Community, and the American Dream* [91]. In this work, Calthorpe defined TOD as a neo-traditional approach to community building that emphasizes sustainability and economic viability.

In North America, there are roughly 200 established TODs, including those in high-density metropolitan cities like Washington, D.C., or Baltimore, Maryland [92]. Modern TODs across North America closely adhere to Calthorpe's original urban planning model. This model is characterized by the strategic positioning of developments within urban areas to yield beneficial outcomes for low- and moderate-income communities, specifically by enhancing connectivity, expanding access to employment opportunities, and generating income in historically underserved areas [93]. Although the purpose of a TOD is to promote social equity within communities that have experienced disconnection and disinvestment, scholarly examination has presented a more nuanced perspective.

Researchers Kim [94] and Wan et al. [95] have identified a phenomenon associated with the adverse effects of TODs called transit-induced gentrification (TIG). TIG refers to the process by which communities proximal to transit systems undergo socioeconomic transformation and displacement due to increased property investments and transportation infrastructure development [96]. The displacement effect resulting from the gentrification of TODs significantly undermines the purported proposed objective of enhancing social equity and connectivity by integrating land use with transportation infrastructure. This contradiction is particularly pronounced when considering the primary demographic of transit users, which is predominately comprised of minorities and individuals with lower socioeconomic backgrounds [97]. This demographic composition is corroborated by a 2001 National Household Travel Survey (NHTS), which indicated that approximately 63% of transit riders in the United States are Black, Hispanic, and from low-income populations. The displacement of frequent transit users from low-income and minority communities' results in a marginal decrease in transit system ridership, as the influx of affluent residents in gentrified TODs typically has access to or prefers alternative modes of transportation, such as walking, cycling, and private vehicles [98]. While there is no definitive solution to deterring TIG, there have been efforts to alleviate some of the associated challenges. For example, metropolitan planning agencies nationwide have shown an increased interest in addressing TIG-related issues by promoting the development of income-restricted housing units near transit stations [96]. This approach to preserving affordable housing for those

within the low-to-moderate socioeconomic sectors aims to facilitate a more equitable distribution of transit access and associated benefits across different demographic groups.

5.3. Vision Zero

Vision Zero is a long-term strategic initiative focused on optimizing the design and operation of road transportation systems to eliminate the occurrence of traffic-related injuries and fatalities [99]. In 1997, the Swedish parliament formally adopted Vision Zero as a comprehensive initiative to enhance road traffic safety. The efficacy of this approach has been substantiated by significant reductions in road fatalities, with overall fatalities cut by 50% and those among car users decreasing by 60% from 2000 to 2010. As a result, Sweden has emerged as one of the leading countries in maintaining and ensuring traffic safety for all its residents [100].

While Vision Zero has demonstrated effectiveness in enhancing road traffic safety, the policy has not been excluded from critique. The most serious of these criticisms leveled against Vision Zero policies concern their alleged role in perpetuating systemic racism and discrimination against racial and ethnic minority communities [101]. Lee [102] argued that the Vision Zero initiative is designed to support a system that perpetuates racial disparities. The initiative relies heavily on policing and surveillance in minority communities to create an environment that affluent white residents perceive as “safe.” A concrete example of this disparity is evident in the discretionary application of stop-and-frisk tactics within traffic safety enforcement. According to a study examining the prevalence of stop-and-frisk experiences in New York City (NYC), it was revealed that 43% of participants reported such encounters. Notably, the incidence was markedly higher among Black (47%) and Hispanic/Latinx (45%) participants compared to their White counterparts (38%) [103]. These stop-and-frisks, as one part of the Vision Zero initiative, are often referred to by legal researchers as the “broken windows approach” or “broken taillight policing” in traffic safety enforcement. This strategy emphasizes the targeting of minor offenses with the belief that such measures will deter individuals from engaging in reckless driving or other deviant activities. However, this practice has resulted in an erosion of public trust in law enforcement due to the disproportionate application of aggressive traffic safety tactics on racial and ethnic minority communities [104].

To address this issue Conner [104] cogently argued that proponents of Vision Zero should prioritize the design and implementation of strategies that promote equity while simultaneously ensuring that current safety initiatives do not result in a disproportionate distribution of benefits and burdens across communities. This approach is aimed at rectifying the unintended consequences of Vision Zero policies by restoring confidence and trust in traffic safety measures and personnel across all demographic groups.

6. Factors for Developing Equity in Transportation Planning

As identified by Oswald Beiler and Mohammed [105], developing and understanding equitable transportation requires an emphasis on the following three critical factors: (a) demographic analysis, (b) socioeconomic considerations, and (c) the integration of transportation systems and land use planning.

- (a) **Demographic Analysis:** The analysis of population parameters, such as age, race, gender, English proficiency, and household composition, provides transportation professionals with the ability to identify equitable and effective solutions and strategies that meet the most pressing needs of diverse communities.
- (b) **Socioeconomic Considerations:** Recognizing the challenges confronted by specific socioeconomic groups, particularly in accessing affordable and reliable transit options,

can lead to planning efforts that better connect individuals to essential services, employment opportunities, and educational resources.

- (c) **Transportation Systems and Land Use:** Integrating transportation systems with land use planning strategically influences how resources are allocated and improves connectivity within communities. This approach ensures that transportation networks are developed in congruence with urban planning initiatives, culminating in more efficient and accessible mobility systems.

The integration of these factors is essential in developing both unimodal and multi-modal transportation approaches that ensure that equity measures are incorporated into every aspect of the transportation process. By considering these factors in transportation programs, policymakers, planners, and researchers can effectively address the needs of communities by creating initiatives that promote equitable access to resources and opportunities, enhance connectivity, and foster economic advancement. This comprehensive approach contributes to the development of a more just and equitable transport network, thereby enabling all individuals to benefit from enhanced mobility systems.

7. Conclusions and Future Research

Equity incorporation is not a new concept. However, it has only recently been considered for transportation studies. Integrating equity into various stages of transportation studies depends on several factors and criteria that define what is considered equitable. Equity in transportation studies has garnered significant attention, as scholars and policymakers recognize its critical importance. Modern perspectives consider transportation as more than just a means of mobility and infrastructure; it is also a means for achieving social equity and justice. This review of literature examines how transportation policies and systems influence various demographic groups, potentially leading to disparities that can restrict access to essential services like healthcare, education, and employment. Such disparities often perpetuate cycles of poverty and social exclusion. Equity in transportation is assessed through the following two lenses: horizontal equity, which emphasizes fair distribution among individuals or groups with similar needs, and vertical equity, which focuses on resource allocation based on diverse needs and abilities, particularly for marginalized groups. Public transit systems designed with equity in mind can significantly enhance accessibility for underserved populations, with strategies like Transit-Oriented Development (TOD) and Bus Rapid Transit (BRT) systems having the potential to improve access and reduce travel times for disadvantaged communities.

Despite advancements, several challenges persist in integrating equity into transportation studies. These challenges include variations in defining, operationalizing, and quantifying social equity objectives in transportation planning. Practical methods for assessing these objectives are lacking, and comprehensive data on the transportation needs and patterns of disadvantaged groups are insufficient, hindering the development of targeted policies. Current evaluations often capture only snapshots of user experiences and may not fully reflect the long-term or nuanced impacts of service quality on different population groups. In addition, the application of proposed frameworks and metrics remains inconsistent, highlighting the need for standardized techniques to incorporate equity issues into transportation policy and project evaluation. To address these challenges, it is essential to develop clearer, more actionable frameworks and metrics for defining, operationalizing, and measuring equity in transportation planning and policies. Enhancing data collection methods to gather comprehensive information on the transportation needs and patterns of diverse demographic groups is crucial for ensuring targeted and effective interventions. Advanced tools and methodologies capable of capturing the dynamic and long-term impacts of transportation systems on different communities, considering tempo-

ral variations in service availability, are necessary. Standardized methods for incorporating equity considerations into transportation project appraisals will enhance consistency across different contexts. Inclusive decision-making processes that involve underrepresented communities will ensure that their voices are considered and their needs are addressed in transportation planning and policy decisions. Implementing progressive redistribution strategies in policies like congestion pricing will ensure the equitable distribution of benefits across all socioeconomic levels. By addressing these recommendations, future research can contribute to developing more equitable transportation systems, ensuring that all members of society benefit from advancements in mobility and infrastructure.

While a limitation to this study is that only one geographic location is considered, future work could consist of conducting a comparative analysis amongst multiple countries facing similar transportation equity challenges. Transportation inequities vary greatly across geographical and socioeconomic contexts, necessitating the tailoring of proposed remedies to local situations. Strategies that work well in dense urban areas, such as improving last-mile connectivity or boosting service frequency, may not be appropriate for low-density suburban or rural areas, where challenges generally stem from limited route coverage and irregular service. To be effective and widely applicable, equity-focused solutions must be informed by localized data, demographic features, land use patterns, and existing infrastructure. Furthermore, involving community stakeholders and considering aspects like income inequality and past disinvestment can assist in customizing tactics to each area's specific needs. In addition, future work is needed to assess equity incorporation challenges and the feasibility of policy implementation in the transportation sector.

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